AP CALCULUS AB

Unit 1 • Worksheet 1-2A

Worksheet 1-2A • Limits to a Point of a Function

Name: _____ Date: _____

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

$\lim_{x \to 3} \dfrac{x^2 - 9}{x^2 - 2x - 3}$ is

(A) 0

(B) $\frac{3}{2}$

(C) 3

(D) nonexistent

Question 2

NO CALCULATOR

$\lim_{x \to 0} \dfrac{\sqrt{x+9} - 3}{x}$ is

(A) 0

(B) $\frac{1}{3}$

(C) $\frac{1}{6}$

(D) 3

Question 3

NO CALCULATOR

$\lim_{x \to 0} \dfrac{\sin(5x)}{x}$ is

(A) 5

(B) 1

(C) $\frac{1}{5}$

(D) 0

Question 4 NO CALCULATOR

$\lim_{x\to 2^-} \dfrac{|x-2|}{x-2}$ is

(A) 1
(B) 0
(C) -1
(D) nonexistent

Question 5 NO CALCULATOR

$\lim_{x\to 4} \dfrac{x+1}{(x-4)^2}$ is

(A) 0
(B) $-\infty$
(C) $\frac{5}{4}$
(D) ∞

Question 6 NO CALCULATOR

$\lim_{x\to 0} x^2 \cos\dfrac{1}{x}$ is

(A) 1
(B) 0
(C) -1
(D) nonexistent

Question 7 (Free response) NO CALCULATOR

Evaluate each limit, showing your work. (a) $\lim_{x\to 5} \dfrac{x^2-3x-10}{x-5}$ (b) $\lim_{x\to 2} \dfrac{x^3-8}{x-2}$ (c) $\lim_{h\to 0} \dfrac{\frac{1}{2+h}-\frac{1}{2}}{h}$ (d) $\lim_{x\to 9} \dfrac{x-9}{\sqrt{x}-3}$ (e) $\lim_{x\to 0} \dfrac{\sin 2x}{\sin 3x}$

Question 8 (Free response) NO CALCULATOR

Let $f(x) = \begin{cases} x^2-1, & x<3 \\ 2x+2, & x\geq 3. \end{cases}$ Find $\lim_{x\to 3^-} f(x)$, $\lim_{x\to 3^+} f(x)$ and $\lim_{x\to 3} f(x)$.